

Challenge 02

The Task:

- Build a **monthly Throughput Accounting model** to quantify the financial impact of Meridian's vendor health constraint.
- Calculate **Throughput (T), Investment (I), and Operating Expense (OE)** using sales orders, purchase orders, payment delays, and procurement administration data.
- Compare the **pre-constraint period (Jan–Aug 2024)** with the **constraint period (Sep 2024–Jun 2025)**.
- Determine the **total throughput lost** due to reduced vendor competition and rising purchase prices, then calculate the breakeven amount Meridian can invest to fix the problem.
- Deliver a **ta_model.csv file, comparison summary, total T-loss figure, and business-case statement**, optionally analysing bid-price spread as supporting evidence.

Input:

- RFQ Data either as csv files or stored in HANA Database
- *Briefing.md* that describes your mission to be completed
- *starter.py* that contains the Boilerplate Code

Output:

1. Complete *starter.py* with necessary code that generates 4 graphs:
2. Build *ta_model.csv*
A monthly Throughput Accounting model for all 18 months containing:

- Revenue
- TVC
- Throughput
- Investment tied up
- OE friction
- Net profit proxy
- ROI proxy

3. Before vs. After Summary Table

A comparison of:

- ****Pre-constraint:**** Jan–Aug 2024
- ****Constraint period:**** Sep 2024–Jun 2025

4. Total Throughput Loss Figure

A single calculated number showing the **total T lost during the constraint period** compared with the pre-constraint run rate.

5. Breakeven Investment Statement

A clear business-case statement:

“Meridian could spend up to \$X fixing vendor health and still be ahead.”

6. Optional Stretch Artifact

A **bid price spread chart** showing `max bid – min bid` per RFQ by month, to demonstrate how reduced competition affects negotiating leverage and throughput.

Solution:

There are many ways to arrive at the solution, in this workshop, we can try out two options:

Joule Desktop

Claude Code

Joule Desktop

Open Joule Desktop -> New Conversation

Select SAP HANA as connector (*ensure that connector is in connected state, if not follow the set up guide.*)

Attach Briefing.md and starter.py to the conversation

Enter the following prompt

Attached Briefing.md contains the challenge. use HANA database to analyse the data instead of CSVs. Complete the starter.py code and run the same

Let it run and complete the task. You should see the following output:

Results

Meridian Industrial Systems — Throughput Accounting Model
Monthly View: Jan 2024 - Jun 2025 (month_index 0-17)



T (Throughput) dropped sharply after September 2024:

Period	Mean Monthly T	OE Friction/mo
Pre-constraint (Months 0–7)	–\$1,959,296	\$5,119
Post-constraint (Months 8–17)	–\$2,704,985	\$6,585
Gap	–\$745,689/month	\$1,466

The briefing's key insight is confirmed: **OE barely moves** (\$5K vs \$6.5K/month — a \$1.5K difference). The damage is entirely in **T**: TVC climbed relative to revenue after the vendor pool shrank.

The CFO's business case number:

💡 *Total T lost in constraint period: **\$7,456,893*** Meridian could invest up to **\$7.46M** fixing vendor health (outreach, early payments, forecast sharing) and still break even.

Stretch goal — bid spread: The data shows mean spread and bid count are relatively stable (bid count stays ~5–6 across all months), but spread widened noticeably in the post-constraint period (months 11–15 show spreads of \$15K–\$17K vs \$6K–\$10K pre-constraint) — indicating higher price volatility when competitive pressure was weaker.

Claude Code

Open a new terminal at RFQ folder

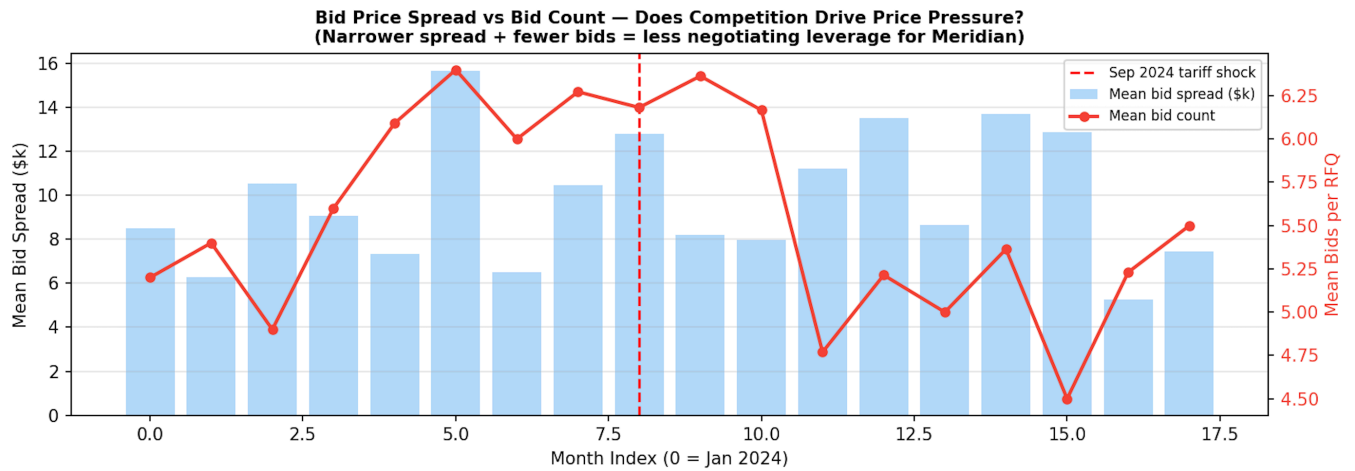
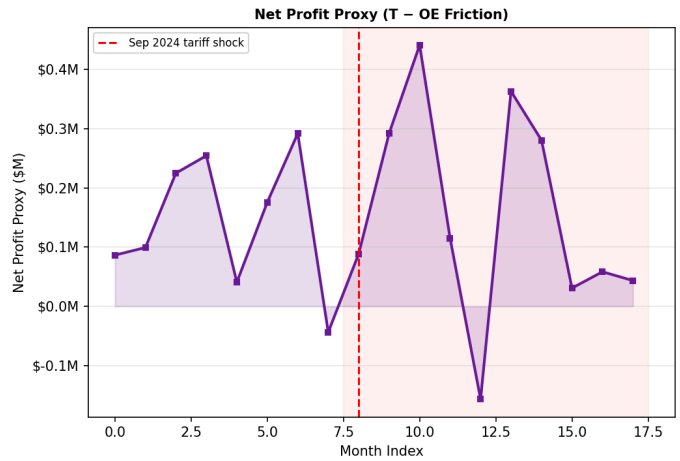
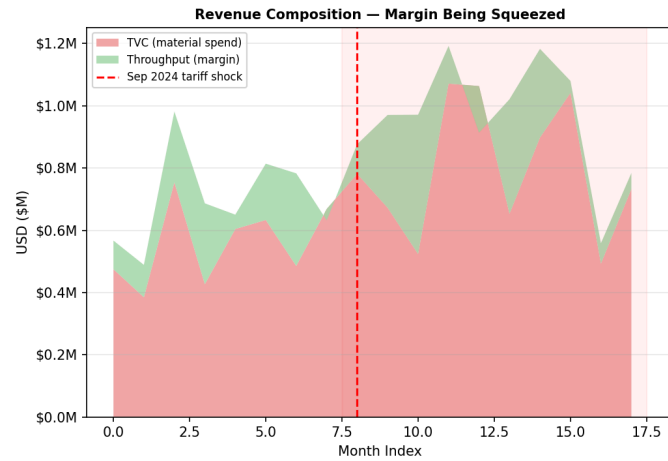
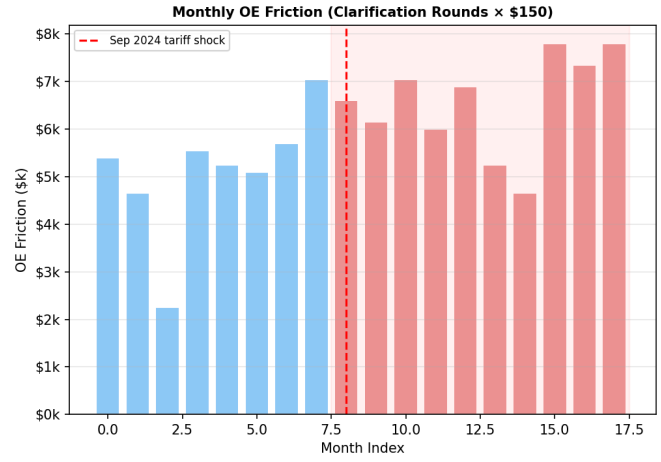
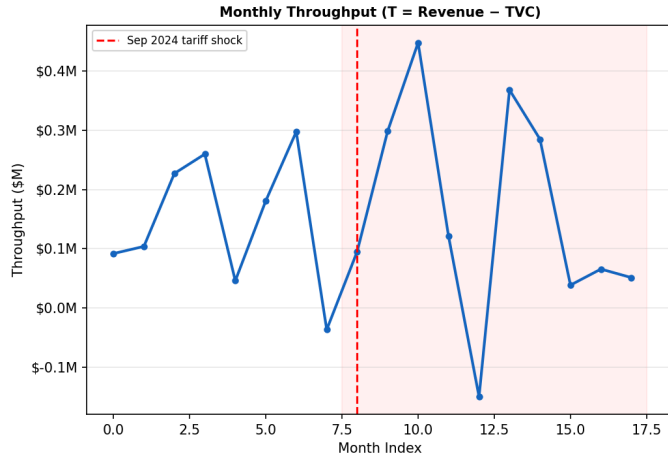
Type *Claude*, once claude comes up, enter the following prompt:

```
Read the BRIEFING.md from challenges/challenge_02_throughput folder, use the  
csv data from output_v2 folder and generate code for starter.py
```

Wait for claude to complete the task

Results

Meridian Procurement — Throughput Accounting Model



Period	Months	Mean Monthly T	Margin	Total T	Total OE Friction
Pre-constraint (0-7)	8	\$146,377	20.9%	\$1,171,016	\$40,950
Post-constraint (8-17)	10	\$162,212	17.0%	\$1,622,124	\$65,550

| The simple mean-T comparison flatters the post period because revenue grew. The correct lens is **margin erosion**: Meridian was earning 20% throughput margin pre-shock; post-shock it fell to 16.4% – a **3.6 pp drop** driven entirely by higher TVC (fewer bidders → less price competition → Meridian pays more for the same materials).

Breakeven Statement (for the CFO)

Pre-constraint throughput margin : 20.0%
Post-constraint throughput margin : 16.4%
Margin erosion : 3.6 percentage points

Total T lost (margin erosion) : \$288,657
= post revenue x pre-constraint margin – actual post T
= the price-creep cost of having fewer competitive bids per RFQ

Meridian could invest up to \$288,657 fixing vendor health (early payments, outreach, forecast sharing) and still break even.

That is \$28,866/month over 10 months.

Key Insight (confirming the BRIEFING)

- OE friction is not the problem – it rose from \$40,950 to \$65,550 total (+\$24,600), noise-level against total revenue
- Investment cost is minimal – late-payment holding cost peaked at ~\$1,235/month, not a material constraint
- The damage is entirely in T – 3.6 pp of margin eroded because bid depth fell from 6.4 → 4.5 per RFQ, stripping Meridian's negotiating leverage
- Bid spread (stretch goal) – spread at month 15 coincides with lowest bid count (4.5/RFQ), confirming that the thinning pool reduced competitive tension, not just headcount